

Sketched Empirical-NTK Capability Fingerprints

Predict

Fine-Tuning Transfer and Interference in Large Language Models

Abstract

Fine-tuning a large language model on new data routinely improves one capability while silently degrading others, but which capabilities will transfer and which will interfere is usually discovered only after the (expensive) training run. We introduce a *capability fingerprint*: for each capability we CountSketch-compress the per-prompt gradients of the answer cross-entropy with respect to a small, fixed parameter subset ($\sim 9\text{M}$ of 7.6B parameters), and predict capability-level transfer from the mean sketch inner product between a candidate fine-tuning set and any held-out capability—*before any training is run*. The predictor is the empirical neural tangent kernel: to first order, one gradient step on capability i changes the loss on capability j by $-\eta \langle g_i, g_j \rangle$. Across four settings validated with one recipe and no per-setting tuning—homogeneous long-context tasks (RULER), heterogeneous real skills (math CoT, code, QA, retrieval, aggregation), a trainer whose updated parameters are *disjoint* from the sketched subset (LoRA on all layers), and a second model family (Llama-3.1-8B)—the off-diagonal Spearman correlation between predicted kernel and measured ΔCE is 0.68, 0.87, 0.62, and 0.69 (all $p \leq 0.004$). An ablation isolates what carries the signal: gradient *direction* is necessary, magnitude alone fails in three of four settings, and representation (embedding) similarity—the standard “similar data” heuristic for data selection—fails in all four and is significantly *anti*-correlated in the heterogeneous setting ($\rho = -0.62$). We also report an honest null: the self-gradient norm as a test-time correctness signal is confidence-in-disguise (cross-validated ΔAUC over free logit baselines $\approx -0.002/-0.0001$). The fingerprint costs one backward pass and 4096 stored floats per prompt, enabling data selection, rehearsal targeting, and forgetting-risk assessment at negligible cost.

1 Introduction

Fine-tuning is the dominant way to adapt a pretrained language model to a new task or distribution, but it is a blunt instrument: gradient steps that improve the target skill also perturb every other capability the model had, sometimes catastrophically. Practitioners mitigate this with data selection, rehearsal of old data, and conservative learning rates, but these decisions are made largely blind—the actual transfer and interference structure is revealed only *after* a training run, when the affected capabilities are re-evaluated. A cheap, *pre*-training-time estimate of “if I train on this data, which capabilities improve, which are damaged, and how strongly” would directly inform data selection, tell us which capabilities to rehearse, and quantify forgetting risk before committing compute.

First-order learning dynamics give a principled handle on this question. A single gradient step on a sample changes the loss on any other sample by the inner product of their loss gradients: the empirical neural tangent kernel (eNTK) [19, 30]. Aggregated over a training set and an evaluation set, this inner product is a first-order prediction of capability-level transfer. The obstacle is cost: full-parameter gradients of a 7–8B model are enormous, and materializing them for every pairing of capabilities is infeasible. We remove the obstacle with two choices—restrict gradients to a small fixed parameter subset, and compress them with a single CountSketch projection [5]—and show empirically that the resulting low-dimensional *fingerprint* preserves enough kernel geometry to predict measured fine-tuning outcomes across model families and even when the trained parameters differ from the sketched ones.

Our starting point was a diagnostic study of long-context “rank demand”: we asked whether the spectral structure that different long-context tasks induce in attention and hidden states predicts correctness. It does not (Section 5, ablation): rank is task-intrinsic and model-stable but adds no per-sample predictive value once entropy is controlled. In the same data, per-prompt gradient (eNTK) features were the strongest signal. This negative→positive contrast pointed us away from activation space and toward gradient space, where first-order training dynamics actually live.

Contributions.

1. **A cheap, pre-training-time predictor of fine-tuning transfer and interference.** Per capability we compute a fingerprint—CountSketch-compressed per-prompt answer-loss gradients on a fixed $\sim 9\text{M}$ -parameter subset (0.12% of the model). The mean sketch inner product between a fine-tuning set and a held-out capability predicts that capability’s measured ΔCE after actual fine-tuning.
2. **Validation across four settings** with the same recipe and no per-setting tuning: homogeneous long-context (RULER), heterogeneous real capabilities, a LoRA trainer whose updated parameters are disjoint from the sketched subset, and a second model family (Llama-3.1-8B). Off-diagonal Spearman between predicted kernel and measured ΔCE : 0.68, 0.87, 0.62, 0.69 (all $p \leq 0.004$).
3. **An ablation isolating the ingredient that carries transfer structure.** Gradient magnitude alone fails in three of four settings; embedding similarity fails in four of four and is significantly anti-correlated in the heterogeneous setting ($\rho = -0.62$). Gradient *direction* is necessary; the full inner product is best everywhere.
4. **A motivating rank-demand diagnostic** showing that attention/hidden-state spectral structure is task-intrinsic and model-stable but has no incremental per-sample predictive value for correctness, whereas per-prompt gradient features do—the useful geometry lives in gradient space.
5. **An honest null.** The self-gradient norm as a test-time correctness signal adds nothing over free confidence baselines (CV $\Delta\text{AUC} \approx -0.002$); raw gradient norms are confidence in disguise, and it is the *kernel* structure, not the norm, that carries training-relevant information.

2 Related Work

NTK and linearized fine-tuning. The neural tangent kernel [19] shows that sufficiently wide networks train as their first-order Taylor expansion about initialization, with dynamics governed by a fixed gradient-inner-product kernel [26, 2]; the *empirical* NTK measured at finite width is a widely used practical surrogate, and fast approximations make it computable for large models [33, 31, 11]. Closest to us, Malladi et al. [30] extend this lens to *fine-tuning* pretrained language models, arguing (and empirically verifying, including an extension to Adam via Tensor Programs) that fine-tuning frequently behaves kernel-like, and a recent theoretical line makes the linearization argument explicit for LLMs [36]. These works establish *that* an eNTK describes LLM fine-tuning on a single task. We take the complementary, constructive step: we *use* the eNTK predictively *across* capabilities, turning the first-order transfer identity $\Delta L(s_j) \approx -\eta \langle g(s_i), g(s_j) \rangle$ into a pre-training-time forecast of which held-out capabilities a candidate fine-tuning set will improve or damage, validated against *measured* ΔCE at 7–8B scale.

Gradient-based data attribution and influence. Influence functions attribute a model’s behavior to individual training points via gradient and inverse-Hessian information [23], with scalable variants that trace gradient descent [35], exploit random projections of per-example gradients [34], or approximate the inverse Hessian for LLM-scale valuation [13, 6, 25, 18]. This literature is *backward-looking*: it explains the predictions of a model that has *already* been trained, and typically requires a per-query solve or Hessian approximation. Our object differs in three ways: (i) we predict the effect of *future* fine-tuning from *base-model* gradients, before any training; (ii) we need no Hessian inversion or optimizer trajectory—a single fixed CountSketch per prompt supports all future capability pairings; and (iii) our unit of prediction is a *capability distribution*, not an individual training point, yielding a full transfer/interference matrix.

Gradient-based data selection and transferability estimation. The machinery closest to ours appears in gradient-based data selection. LESS [47] selects instruction-tuning data by projecting optimizer-aware (Adam) gradients collected *along a training trajectory* and matching them to a *fixed target task*; NTK-Selector [44] concurrently uses random-projected NTK scores at initialization to mine auxiliary data for a *single* low-resource domain. We share the low-dimensional gradient-feature idea but differ in goal and validation: rather than ranking data for one target, we predict the *entire* cross-capability transfer *and* interference matrix from base-model gradients and check it against measured ΔCE , including the case where the trained parameters (all-layer LoRA) are *disjoint* from the sketched subset. A parallel thread estimates *task affinity* from gradients for multi-task grouping—TAG measures how one task’s gradient step changes another’s loss [10], Grad-TAG fits a logistic surrogate on projected gradients to predict combined-task loss [28], and earlier work groups tasks by cooperation/competition [40]—but these target *which tasks to co-train* (positive transfer for grouping), operate during or across training runs, and are demonstrated below LLM scale. Transferability metrics such as Task2Vec [1], TaskEmb [43], LEEP [32], LogME [49], and the H-score [4, 51] likewise *embed* tasks or score source-model reusability; notably Task2Vec fingerprints a task with Fisher-weighted gradients, but for meta-learning similarity—it does not predict a measured fine-tuning ΔCE transfer matrix, and uses a different gradient object than our pairwise eNTK inner products.

Forgetting and task interference. Fine-tuning LLMs degrades pretraining capabilities [29, 24], extending classic studies of forgetting [22, 42], and is commonly mitigated by regularization or rehearsal/replay [39, 16]. In weight space, task vectors interfere when models are merged, motivating sign- and magnitude-based conflict resolution [17, 48]; in gradient space, multi-task optimization identifies conflicting (negatively aligned) gradients and surgically removes the conflict [50, 45, 9]. The work theoretically nearest to our interference claim is Doan et al. [8], who show that catastrophic forgetting between two tasks grows with their NTK overlap and formalize task similarity through an NTK overlap matrix, complemented by representational analyses of what drives forgetting [38]. That line is theoretical, is developed for two-task continual learning in vision, and prescribes an *intervention* (orthogonal gradient descent) rather than a validated quantitative forecast. We instead *measure* the full signed transfer/interference matrix induced by real fine-tuning of a 7–8B LLM and show that a cheap sketched-eNTK kernel predicts it—capturing both damage (positive ΔCE) and benefit (negative ΔCE), across heterogeneous real capabilities and two model families—which in turn enables rehearsal targeting. This complements our companion function-space analysis of when such forgetting is low-rank [14].

Sketching gradients. We compress each gradient with CountSketch [5], an unbiased inner-product-preserving projection in the Johnson–Lindenstrauss family [20] also known as the hashing trick [46]; the same random-projection principle underlies gradient attribution [34] and selection [47]. Our use is distinctive in that one fixed sketch per prompt is reused across *all* capability pairings to reconstruct the Gram matrix without ever materializing gradients. Finally, we report an honest negative result: the *norm* of a self-generated answer’s gradient—a natural test-time confidence signal in the spirit of gradient-based uncertainty [27]—adds nothing over free logit-based baselines, underscoring that the predictive content lives in the *inner-product structure* of the kernel, not in gradient magnitude.

3 Mathematical Background

Teacher-forced answer loss and per-sample gradient. Let f_θ be a causal language model with parameters θ . A *capability* c is a distribution over (x, y) pairs, where x is a prompt and $y = (y_1, \dots, y_T)$ its reference answer. For a sample $s = (x, y)$ the teacher-forced answer loss is the cross-entropy of y under the model conditioned on x ,

$$L(s; \theta) = -\frac{1}{T} \sum_{t=1}^T \log f_\theta(y_t \mid x, y_{<t}), \quad (1)$$

computed only over answer tokens (the prompt is masked). We restrict attention to a fixed parameter subset $S \subseteq \theta$ and define the per-sample gradient $g(s) = \nabla_{\theta_S} L(s; \theta) \in \mathbb{R}^{|S|}$.

First-order effect of one gradient step. Consider one gradient-descent step of size η taken on sample s_i , $\theta_S \leftarrow \theta_S - \eta g(s_i)$. A first-order Taylor expansion of the loss on a second sample s_j around θ_S gives

$$\Delta L(s_j) = L(s_j; \theta - \eta g(s_i)) - L(s_j; \theta) = -\eta \langle g(s_i), g(s_j) \rangle + O(\eta^2). \quad (2)$$

Averaging (2) over a training set T_i (drawn from capability i) and an evaluation set E_j (from capability j) yields a first-order prediction of capability-level transfer. Define the predicted transfer matrix

$$K[i, j] = \frac{1}{|T_i||E_j|} \sum_{s \in T_i} \sum_{s' \in E_j} \langle g(s), g(s') \rangle, \quad \text{predicted } \Delta \text{CE}[i, j] \propto -K[i, j]. \quad (3)$$

Large positive $K[i, j]$ predicts that training on i *reduces* the loss on j (transfer); negative $K[i, j]$ predicts an *increase* (interference). This is a prediction of direction and relative magnitude, not of the exact loss change: the constant of proportionality absorbs η , the number of steps, and higher-order curvature, so we evaluate it by rank (Spearman) and linear (Pearson) agreement with the measured ΔCE .

Empirical neural tangent kernel. The matrix $\langle g(s), g(s') \rangle$ is the empirical NTK of f_θ evaluated on the loss gradients at the current parameters [19, 30]. Unlike the infinite-width NTK it is not constant, but for the short fine-tuning horizons of interest the first-order regime of (2) is the relevant one, and the eNTK Gram matrix over capability samples is exactly the object that governs it. Our method is a scalable estimator of this Gram matrix and a study of how well its structure predicts real fine-tuning outcomes.

CountSketch and inner-product preservation. Materializing $g(s) \in \mathbb{R}^{|S|}$ for every sample is wasteful when only inner products are needed. CountSketch [5] is a sparse random projection $\Phi : \mathbb{R}^{|S|} \rightarrow \mathbb{R}^m$ defined by two fixed hash functions: a bucket $h : \{1, \dots, |S|\} \rightarrow \{1, \dots, m\}$ and a sign $\sigma : \{1, \dots, |S|\} \rightarrow \{-1, +1\}$, with $(\Phi g)_b = \sum_{k: h(k)=b} \sigma(k) g_k$. Drawn once (fixed seed) and shared across all samples, Φ is an unbiased estimator of inner products,

$$\mathbb{E}[\langle \Phi g, \Phi g' \rangle] = \langle g, g' \rangle, \quad \text{Var}[\langle \Phi g, \Phi g' \rangle] = O\left(\frac{1}{m} \|g\|^2 \|g'\|^2\right), \quad (4)$$

the sign flips cancelling cross terms in expectation and the variance decaying as $1/m$. With $m = 4096$ the standard deviation of a normalized inner product is $O(1/\sqrt{m}) \approx 1.6\%$; because our Gram entries are *averaged* over $|T_i| \times |E_j| \geq 30 \times 30$ pairs, the per-cell error is suppressed by a further factor of $\sqrt{|T_i||E_j|}$, which is why $m = 4096$ suffices to recover the capability-level kernel structure at 16 KB per prompt.

Why a parameter subset still captures kernel structure. Restricting S to a small set of tensors loses parameters but not, empirically, the geometry that matters: gradient directions are strongly shared across the network, so a subset’s Gram matrix is a good proxy for the full one. Section 5 makes this concrete: in the LoRA setting the trained parameters (low-rank adapters on *all* layers) are disjoint from the sketched subset (q/v projections of five layers), yet the subset fingerprint still predicts the measured transfer, evidence that we are reading a network-wide gradient geometry rather than memorizing exactly what we train.

4 Method

Parameter subset. S is the query and value projections (`q.proj`, `v.proj`) of five layers at depth fractions $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$ —approximately 9M parameters of a 7.6B model (0.12%). It is fixed once per model and never tuned per task.

Sketched fingerprints. For each sample we run one forward and one backward pass on the frozen model with gradients enabled only on S , apply Φ per parameter tensor and sum, and store the resulting $m = 4096$ -dimensional sketch. The fingerprint of a capability c is the set $\{\Phi g(s) : s \in c\}$, 16 KB per prompt. No training is performed and no labels beyond the reference answers already needed for evaluation are used. With gradient checkpointing the procedure fits generation-scale memory.

Table 1: Validation settings. All use $n_{\text{train}} = 32$, $n_{\text{eval}} = 30$ per capability, 3 epochs; outcome is gold-answer ΔCE .

Setting	What it tests	Trainer	Capabilities
RULER 5×5	base validity	Adam on subset S	5 RULER families @1024
Heterogeneous	real, dissimilar skills	Adam on subset S	GSM8K CoT, MBPP, SQuAD QA, needle, aggregation
LoRA	trained $\neq$ sketched params	rank-16 LoRA, all q/v layers	5 RULER families
Llama-3.1-8B	model-family generality	Adam on subset S	5 RULER families

Table 2: Main result: correlation of the predicted kernel ($-K$) with measured ΔCE , for the **ntk_kernel** predictor. “All” = 25 pairs; “off-diag” = 20 cross-capability pairs (stricter). p -values as reported; $< 10^{-4}$ shown for machine-zero.

	Spearman (all)	Pearson (all)	Spearman (off-diag)	p (off-diag)
RULER (subset)	0.79	0.98	0.68	0.001
Heterogeneous	0.91	0.92	0.87	$< 10^{-4}$
LoRA	0.78	0.88	0.62	0.004
Llama-3.1-8B	0.82	0.75	0.69	0.0007

Prediction protocol. Given capabilities $c_1, \dots, c_n$ with seed-disjoint train sets T_i and eval sets E_j : (1) sketch all train and eval samples once with the same Φ ; (2) form the predicted transfer matrix K via (3) from sketch inner products; (3) for ground truth, fine-tune the model on T_i alone, measure ΔCE on every E_j , restore weights, and repeat for each i ; (4) score by Spearman and Pearson correlation between $-K$ and measured ΔCE over all (i, j) pairs, and—the stricter test—over *off-diagonal pairs only*, which removes the easy “training on X helps X ” diagonal signal and isolates cross-capability transfer/interference.

Validation settings. We validate the same recipe on four axes (Table 1). The LoRA setting is the key robustness axis: the fingerprint is computed on five layers’ q/v while training updates rank-16 LoRA adapters on the q/v of *all* layers, so the prediction can only succeed if gradient geometry is shared across the network rather than because we sketch exactly what we train.

Trainer details. Fine-tuning fp16 weights directly with Adam [21] destroys the model; we keep fp32 master copies of the trainable parameters, apply dynamic loss scaling, and skip any update whose gradients are non-finite (Appendix A). Learning rate is 2×10^{-5} for subset training and 2×10^{-4} for LoRA. Capabilities and models are drawn from RULER [15], GSM8K [7], MBPP [3], SQuAD [37], Qwen2.5-7B-Instruct [41], and Llama-3.1-8B-Instruct [12].

Baselines. Against the same measured ΔCE matrices we compare four pre-training predictors: **ntk_kernel** (ours, mean $\langle g, g' \rangle$, direction $\times$ magnitude); **ntk_cos** (mean cosine, direction only); **grad_magnitude** (mean $\|g\| \|g'\|$, magnitude only); and **embed_cos** (mean cosine of last-hidden-state mean-pooled prompt embeddings—the standard representation-similarity “similar data” heuristic used in embedding-based data selection).

5 Results

The kernel predicts measured transfer in all four settings. Table 2 reports, for our predictor, the Spearman and Pearson correlation between $-K$ and measured ΔCE over all 25 (i, j) pairs, and the off-diagonal Spearman over the 20 cross-capability pairs. Off-diagonal agreement is 0.68/0.87/0.62/0.69 (RULER/hetero/LoRA/Llama), all significant at $p \leq 0.004$; pooled Pearson over all pairs is 0.98/0.92/0.88/0.75. Figure 1 shows the corresponding scatter plots, with off-diagonal points highlighted: the predicted ordering tracks the measured ΔCE across more than three orders of magnitude of kernel scale, and—critically—does so on the off-diagonal, where the diagonal “self-transfer” shortcut is removed.

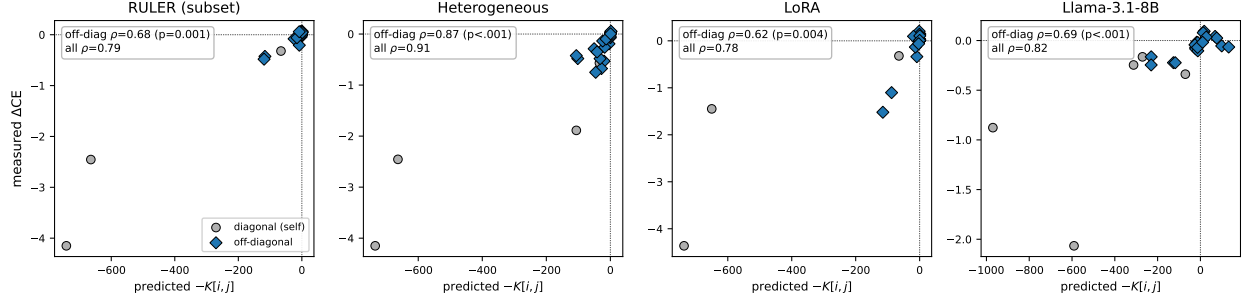


Figure 1: Predicted kernel ($-K[i, j]$, x) vs. measured ΔCE (y) across the four settings. Grey circles are diagonal (self) pairs, blue diamonds off-diagonal (cross-capability) pairs. Off-diagonal Spearman ρ (the stricter score) and all-pairs ρ annotated. The predicted kernel orders the measured loss changes across the full range, including the off-diagonal where the self-transfer shortcut is absent.

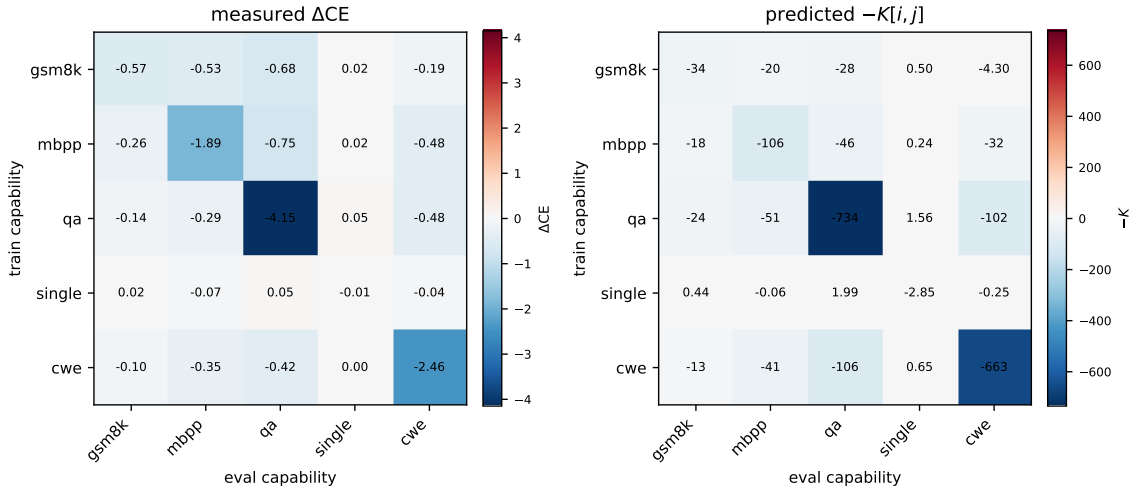


Figure 2: Heterogeneous setting: measured ΔCE (left) and predicted $-K$ (right), same row/column order (train capability $\times$ eval capability), diverging colormap centered at zero. The block structure matches: strong mutual transfer among GSM8K/MBPP/QA, near-zero coupling to needle retrieval, large negative diagonal.

Predicted and measured matrices agree cell by cell. Figure 2 places the measured ΔCE matrix beside the predicted $-K$ matrix for the heterogeneous setting under a common row/column order and a diverging colormap centered at zero. The two share the same block structure: large negative diagonal (self-training helps), strong transfer among the language capabilities (GSM8K, MBPP, QA mutually reduce loss), and near-zero coupling to the retrieval capability (needle), whose row and column are close to zero in both matrices. This heterogeneous setting is where the method is most useful and also where it is strongest (0.87 off-diagonal), because real capabilities span the widest range of gradient geometry.

Protocol note. All ΔCE values are gold-answer cross-entropy changes measured by fine-tuning on a single capability’s train set and re-evaluating every capability (Section 4). Accuracy deltas are tracked as a secondary outcome but are noisy at $n_{\text{eval}} = 30$ (Appendix C); we therefore report ΔCE , the smoother first-order quantity that the theory predicts.

Table 3: Baseline shootout: off-diagonal Spearman ρ vs. measured ΔCE (p in parentheses; $n.s.$ = $p > 0.05$; “—” = no data). Direction-carrying predictors (top two rows) dominate; magnitude and embedding similarity fail in most settings.

Predictor	RULER	Heterogeneous	LoRA	Llama
ntk_kernel (ours)	0.68 (.001)	0.87 ($< 10^{-4}$)	0.62 (.004)	0.69 (.0007)
ntk_cos	0.67 (.001)	0.79 ($< 10^{-4}$)	0.55 (.012)	0.59 (.006)
grad_magnitude	0.25 (<i>n.s.</i>)	0.80 ($< 10^{-4}$)	0.28 (<i>n.s.</i>)	0.09 (<i>n.s.</i>)
embed_cos	—	−0.62 (.003)	−0.30 (<i>n.s.</i>)	0.02 (<i>n.s.</i>)

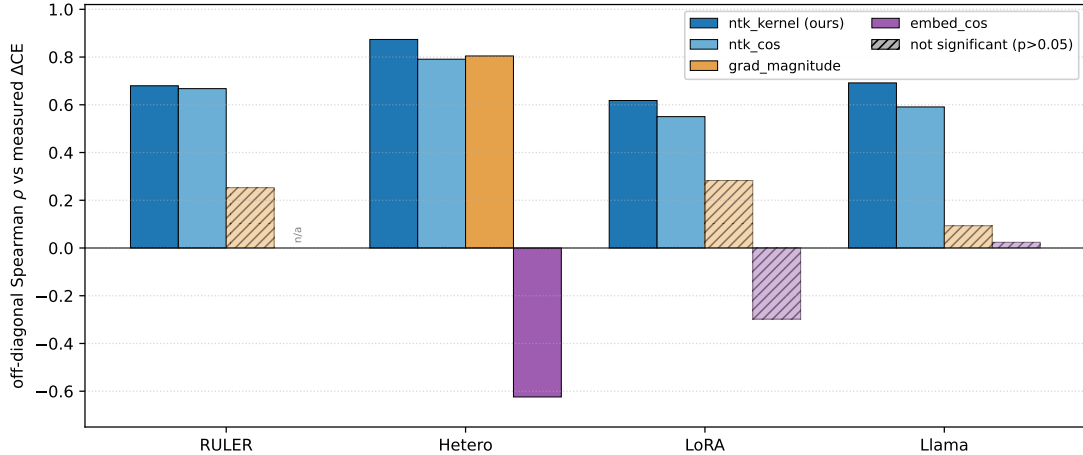


Figure 3: Off-diagonal Spearman per predictor across the four settings. Hatched, translucent bars are not significant ($p > 0.05$); “n/a” marks the missing RULER embedding baseline. The inner-product and cosine predictors (blue) are the only ones that are significant everywhere; embedding similarity (purple) is negative where it is significant.

6 Ablation

What carries the transfer structure? Table 3 and Figure 3 compare the four predictors by off-diagonal Spearman across the four settings. Three findings are consistent. *(i) Direction is necessary.* The full kernel (ntk_kernel) is the best predictor everywhere, with cosine (ntk_cos, direction only) close behind—the two dominate every other predictor in every setting. *(ii) Magnitude alone is not enough.* grad_magnitude collapses to non-significant in three of four settings ($\rho = 0.25, 0.28, 0.09$; $p > 0.05$); it works only in the heterogeneous setting, where capability-scale differences happen to align with transfer. *(iii) Representation similarity fails.* embed_cos—the standard heuristic that “representationally similar data transfers”—is useless or actively misleading: it is non-significant in the LoRA and Llama settings, and significantly *anti*-correlated in the heterogeneous setting ($\rho = -0.62$, $p = 0.003$). Representationally similar prompts do not predict training-time transfer; the gradient inner product does. (The RULER embedding baseline is omitted: embeddings were not stored for that run.)

LoRA vs. subset: trained $\neq$ sketched parameters. The LoRA setting is a robustness test, not merely a fourth dataset: the fingerprint is sketched on the q/v projections of five layers, while training updates low-rank adapters on the q/v of all layers—disjoint parameter sets. The predictor nonetheless attains 0.62 off-diagonal Spearman ($p = 0.004$) and 0.88 pooled Pearson, only modestly below the matched-parameter RULER setting. This supports the claim of Section 3: the fingerprint reads a network-wide gradient geometry, so the sketched subset need not coincide with the trained parameters.

The self-gradient null. A tempting simpler idea is to use the gradient norm of the model’s *own generated* answer as a test-time correctness signal. It fails. On $n = 1000$ (Qwen) and 983 (Llama) RULER samples,

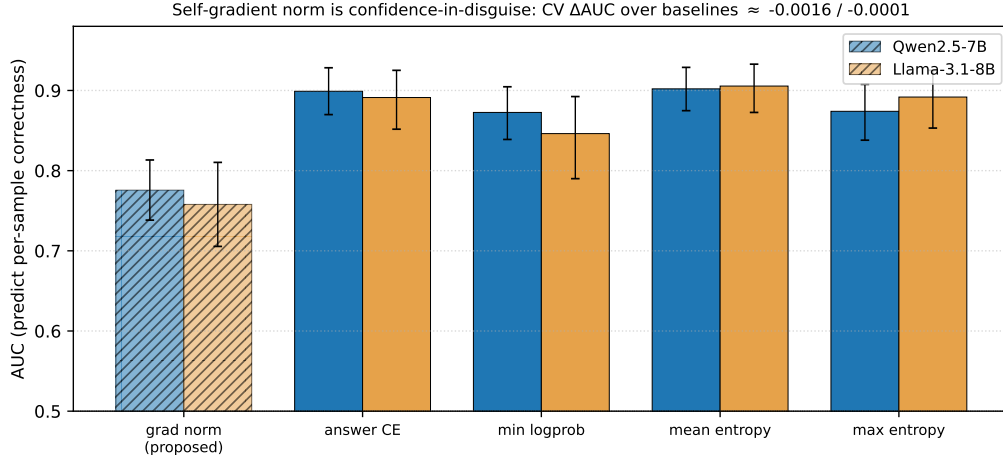


Figure 4: Self-gradient null. Correctness-prediction AUC (with 95% CIs) for the proposed self-gradient norm (hatched) vs. free confidence baselines, on both models. The gradient norm is the weakest predictor and adds cross-validated $\Delta\text{AUC} \approx -0.002 / -0.0001$ on top of the baselines.

self-gradient-norm AUC for predicting correctness is 0.78/0.76, well below free logit baselines: answer cross-entropy 0.90/0.89 and mean-token entropy 0.90/0.91 (Figure 4). Crucially, in a 5-fold cross-validated logistic regression, adding the gradient norm on top of the confidence baselines changes AUC by -0.0016 (Qwen) and -0.0001 (Llama)—no incremental value. Gradient *norms* are confidence in disguise; only the *kernel* (inner-product structure across samples) carries information not already in the logits. This is the cautionary counterpart to the main result: the norm is not the signal, the geometry is.

The rank-demand null (motivation evidence). The study that motivated the method is itself a null that points to gradient space. On five RULER families at context lengths 1k–8k (1000 samples per model), hidden-state effective rank is task-intrinsic and orders the families near-identically across Qwen and Llama (Figure 5), and the ordering is *not* the attention-entropy ordering, so it is not an entropy artifact. Yet in 5-fold cross-validated logistic regression predicting correctness, adding rank features on top of context length, family, and entropy changes AUC by -0.002 (Qwen) and $+0.002$ (Llama)—an in-sample gain of $+0.010$ that vanishes under cross-validation, i.e. pure overfit. In the same runs, per-prompt eNTK features showed exactly the structure activation-space metrics lacked: within-family gradient cosine similarity is 0.21/0.41 (Qwen/Llama) versus 0.02/0.05 across families, and self-kernel norms span hundreds-fold across families. Prompts from the same capability occupy a tight cone in gradient space while different capabilities are near-orthogonal—the geometry the method exploits.

7 Conclusion

We introduced capability fingerprints—CountSketch-compressed per-prompt answer-loss gradients on a small fixed parameter subset—and showed that the mean fingerprint inner product between a candidate fine-tuning set and a held-out capability predicts the measured post-fine-tuning change in that capability’s loss, before any training is run. Across four settings (homogeneous long-context, heterogeneous real skills, a LoRA trainer with disjoint parameters, and a second model family) the off-diagonal predicted-vs-measured Spearman correlation is 0.68–0.87 (all $p \leq 0.004$). The ablation localizes the signal to gradient *direction*: magnitude alone fails in most settings and embedding similarity fails everywhere, sometimes anti-predicting. The method costs one backward pass and 16 KB per prompt and directly supports data selection, rehearsal targeting, and forgetting-risk assessment.

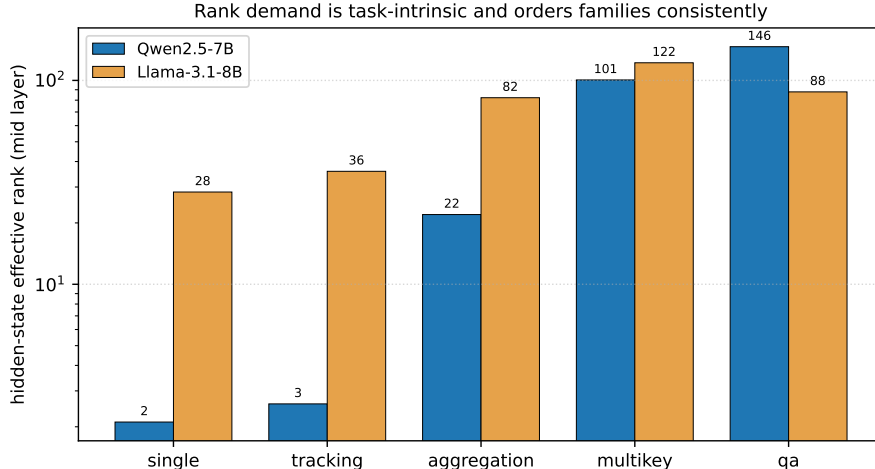


Figure 5: Hidden-state effective rank (mid layer, log scale) by RULER task family for both models. Rank demand is task-intrinsic and orders the families consistently across model families (qa/multikey high, single/tracking low), yet adds no cross-validated per-sample predictive value for correctness—motivating the shift from activation space to gradient space.

Limitations. Our evaluation uses five capabilities per setting and short fine-tuning horizons (3 epochs of 32 samples); first-order theory predicts the linear approximation degrades for longer/higher-learning-rate training, which we have not yet stress-tested. The outcome is ΔCE ; accuracy deltas are noisier at $n_{\text{eval}} = 30$. Experiments are at 7–8B scale on two model families and fp16 Turing hardware, and the parameter subset (q/v of five layers) is fixed a priori—subset-choice sensitivity is untested.

Future work. Natural extensions, currently running, are: capability splits for safety, multilingual, and reasoning behaviors; horizon scaling (does the first-order prediction decay predictably with training length?); and rehearsal targeting (selecting data to rehearse by predicted interference and comparing against random rehearsal). We report these as directions, not results.

References

- [1] Alessandro Achille, Michael Lam, Rahul Tewari, Avinash Ravichandran, Subhansu Maji, Charless Fowlkes, Stefano Soatto, and Pietro Perona. Task2Vec: Task embedding for meta-learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- [2] Sanjeev Arora, Simon S. Du, Wei Hu, Zhiyuan Li, Ruslan Salakhutdinov, and Ruosong Wang. On exact computation with an infinitely wide neural net. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.
- [3] Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. Program synthesis with large language models. *arXiv preprint arXiv:2108.07732*, 2021.
- [4] Yajie Bao, Yang Li, Shao-Lun Huang, Lin Zhang, Lizhong Zheng, Amir Zamir, and Leonidas J. Guibas. An information-theoretic approach to transferability in task transfer learning. In *IEEE International Conference on Image Processing (ICIP)*, 2019.
- [5] Moses Charikar, Kevin Chen, and Martin Farach-Colton. Finding frequent items in data streams. In *International Colloquium on Automata, Languages, and Programming (ICALP)*, pages 693–703, 2002.

- [6] Sang Keun Choe, Hwijeen Ahn, Juhan Bae, Kewen Zhao, Minsoo Kang, Youngseog Chung, Adithya Pratapa, Willie Neiswanger, Emma Strubell, Teruko Mitamura, Jeff Schneider, Eduard Hovy, Roger Grosse, and Eric Xing. What is your data worth to GPT? LLM-scale data valuation with influence functions. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [7] Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- [8] Thang Doan, Mehdi Abbana Bennani, Bogdan Mazouze, Guillaume Rabusseau, and Pierre Alquier. A theoretical analysis of catastrophic forgetting through the NTK overlap matrix. In *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2021.
- [9] Yunshu Du, Wojciech M. Czarnecki, Siddhant M. Jayakumar, Mehrdad Farajtabar, Razvan Pascanu, and Balaji Lakshminarayanan. Adapting auxiliary losses using gradient similarity. *arXiv preprint arXiv:1812.02224*, 2018.
- [10] Christopher Fifty, Ehsan Amid, Zhe Zhao, Tianhe Yu, Rohan Anil, and Chelsea Finn. Efficiently identifying task groupings for multi-task learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 2021.
- [11] Stanislav Fort, Gintare Karolina Dziugaite, Mansheej Paul, Sepideh Kharaghani, Daniel M. Roy, and Surya Ganguli. Deep learning versus kernel learning: An empirical study of loss landscape geometry and the time evolution of the neural tangent kernel. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.
- [12] Aaron Grattafiori, Abhimanyu Dubey, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- [13] Roger Grosse, Juhan Bae, Cem Anil, Nelson Elhage, Alex Tamkin, Amirhossein Tajdini, Benoit Steiner, Dustin Li, Esin Durmus, Ethan Perez, Evan Hubinger, Kamilė Lukošiuūtė, Karina Nguyen, Nicholas Joseph, Sam McCandlish, Jared Kaplan, and Samuel R. Bowman. Studying large language model generalization with influence functions. *arXiv preprint arXiv:2308.03296*, 2023.
- [14] Ido Nitzan Hidekel and Dan Raviv. Catastrophic forgetting is low-rank: A function-space theory for continual adaptation. *arXiv preprint arXiv:2606.18024*, 2026.
- [15] Cheng-Ping Hsieh, Simeng Sun, Samuel Krizan, Shantanu Acharya, Dima Rekeshe, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What’s the real context size of your long-context language models? In *Conference on Language Modeling (COLM)*, 2024.
- [16] Jianheng Huang, Leyang Cui, Ante Wang, Chengyi Yang, Xinting Liao, Linfeng Song, Junfeng Yao, and Jinsong Su. Mitigating catastrophic forgetting in large language models with self-synthesized rehearsal. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.
- [17] Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hananeh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations (ICLR)*, 2023.
- [18] Andrew Ilyas, Sung Min Park, Logan Engstrom, Guillaume Leclerc, and Aleksander Madry. Datamodels: Predicting predictions from training data. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022.
- [19] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

- [20] William B. Johnson and Joram Lindenstrauss. Extensions of Lipschitz mappings into a Hilbert space. *Contemporary Mathematics*, 26:189–206, 1984.
- [21] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.
- [22] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences (PNAS)*, 114(13):3521–3526, 2017.
- [23] Pang Wei Koh and Percy Liang. Understanding black-box predictions via influence functions. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, 2017.
- [24] Suhas Kotha, Jacob Mitchell Springer, and Aditi Raghunathan. Understanding catastrophic forgetting in language models via implicit inference. In *International Conference on Learning Representations (ICLR)*, 2024.
- [25] Yongchan Kwon, Eric Wu, Kevin Wu, and James Zou. DataInf: Efficiently estimating data influence in LoRA-tuned LLMs and diffusion models. In *International Conference on Learning Representations (ICLR)*, 2024.
- [26] Jaehoon Lee, Lechao Xiao, Samuel S. Schoenholz, Yasaman Bahri, Roman Novak, Jascha Sohl-Dickstein, and Jeffrey Pennington. Wide neural networks of any depth evolve as linear models under gradient descent. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.
- [27] Jinsol Lee and Ghassan AlRegib. Gradients as a measure of uncertainty in neural networks. In *IEEE International Conference on Image Processing (ICIP)*, 2020.
- [28] Dongyue Li, Aneesh Sharma, and Hongyang R. Zhang. Scalable multitask learning using gradient-based estimation of task affinity. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2024.
- [29] Yun Luo, Zhen Yang, Fandong Meng, Yafu Li, Jie Zhou, and Yue Zhang. An empirical study of catastrophic forgetting in large language models during continual fine-tuning. *arXiv preprint arXiv:2308.08747*, 2023.
- [30] Sathika Malladi, Alexander Wettig, Dingli Yu, Danqi Chen, and Sanjeev Arora. A kernel-based view of language model fine-tuning. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- [31] Mohamad Amin Mohamadi, Wonho Bae, and Danica J. Sutherland. A fast, well-founded approximation to the empirical neural tangent kernel. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- [32] Cuong V. Nguyen, Tal Hassner, Matthias Seeger, and Cedric Archambeau. LEEP: A new measure to evaluate transferability of learned representations. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.
- [33] Roman Novak, Jascha Sohl-Dickstein, and Samuel S. Schoenholz. Fast finite width neural tangent kernel. In *Proceedings of the 39th International Conference on Machine Learning (ICML)*, 2022.
- [34] Sung Min Park, Kristian Georgiev, Andrew Ilyas, Guillaume Leclerc, and Aleksander Madry. TRAK: Attributing model behavior at scale. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.
- [35] Garima Pruthi, Frederick Liu, Satyen Kale, and Mukund Sundararajan. Estimating training data influence by tracing gradient descent. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.

- [36] Zahra Rahimi Afzal, Tara Esmaeilbeig, Mojtaba Soltanalian, and Mesrob I. Ohannessian. Linearization explains fine-tuning in large language models. *arXiv preprint arXiv:2602.08239*, 2026.
- [37] Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2383–2392, 2016.
- [38] Vinay V. Ramasesh, Ethan Dyer, and Maithra Raghu. Anatomy of catastrophic forgetting: Hidden representations and task semantics. In *International Conference on Learning Representations (ICLR)*, 2021.
- [39] Thomas Scialom, Tuhin Chakrabarty, and Smaranda Muresan. Fine-tuned language models are continual learners. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2022.
- [40] Trevor Standley, Amir R. Zamir, Dawn Chen, Leonidas Guibas, Jitendra Malik, and Silvio Savarese. Which tasks should be learned together in multi-task learning? In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, 2020.
- [41] Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.
- [42] Mariya Toneva, Alessandro Sordoni, Remi Tachet des Combes, Adam Trischler, Yoshua Bengio, and Geoffrey J. Gordon. An empirical study of example forgetting during deep neural network learning. In *International Conference on Learning Representations (ICLR)*, 2019.
- [43] Tu Vu, Tong Wang, Tsendsuren Munkhdalai, Alessandro Sordoni, Adam Trischler, Andrew Mattarella-Micke, Subhransu Maji, and Mohit Iyyer. Exploring and predicting transferability across NLP tasks. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020.
- [44] Pingjie Wang, Hongcheng Liu, Yusheng Liao, Ziqing Fan, Yaxin Du, Shuo Tang, Yanfeng Wang, and Yu Wang. Mining useful general data for low-resource domain adaptation via neural tangent kernels. *arXiv preprint arXiv:2511.07380*, 2026.
- [45] Zirui Wang, Yulia Tsvetkov, Orhan Firat, and Yuan Cao. Gradient vaccine: Investigating and improving multi-task optimization in massively multilingual models. In *International Conference on Learning Representations (ICLR)*, 2021.
- [46] Kilian Weinberger, Anirban Dasgupta, Josh Attenberg, John Langford, and Alex Smola. Feature hashing for large scale multitask learning. In *Proceedings of the 26th International Conference on Machine Learning (ICML)*, 2009.
- [47] Mengzhou Xia, Sadhika Malladi, Suchin Gururangan, Sanjeev Arora, and Danqi Chen. LESS: Selecting influential data for targeted instruction tuning. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024.
- [48] Prateek Yadav, Derek Tam, Leshem Choshen, Colin Raffel, and Mohit Bansal. TIES-Merging: Resolving interference when merging models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.
- [49] Kaichao You, Yong Liu, Jianmin Wang, and Mingsheng Long. LogME: Practical assessment of pre-trained models for transfer learning. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, 2021.
- [50] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, 2020.
- [51] Amir R. Zamir, Alexander Sax, William Shen, Leonidas J. Guibas, Jitendra Malik, and Silvio Savarese. Taskonomy: Disentangling task transfer learning. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.

A Trainer details (fp16 with fp32 masters)

All models are served in fp16 on Quadro RTX 8000 (Turing, compute capability 7.5, 48 GB). Direct fp16 Adam on the trainable parameters diverges (fp16 rounding of the small Adam updates destroys the model). Our trainer keeps an fp32 master copy of every trainable parameter; the forward and backward passes run in fp16 with dynamic loss scaling (scale halved on non-finite gradients, grown otherwise), gradients are unscaled and cast to fp32, the Adam step is taken on the fp32 masters, and the fp16 weights are refreshed from the masters. Any step with non-finite gradients after unscaling is skipped. Subset training uses learning rate 2×10^{-5} ; LoRA (rank 16, $\alpha = 32$, on all q/v projections) uses 2×10^{-4} . Both run 3 epochs over $n_{\text{train}} = 32$ samples per capability. Weights are restored to the pretrained checkpoint between capabilities so each ΔCE row is measured against the same reference model.

B Hyperparameters and evaluation

Per capability: $n_{\text{train}} = 32$, $n_{\text{eval}} = 30$, seed-disjoint. Sketch dimension $m = 4096$ (one CountSketch Φ per model, fixed seed, applied per parameter tensor and summed). Parameter subset S : **q_proj** and **v_proj** at layer-depth fractions $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1\}$, $\approx 9\text{M}$ parameters. Gradients use teacher-forced answer cross-entropy with the prompt masked. RULER context length is 1024 for the interference experiments and $\{1024, 2048, 4096, 8192\}$ for the rank-demand diagnostic. Correlations are Spearman and Pearson with two-sided p -values; the off-diagonal statistic excludes the five diagonal (self) cells.

C Full measured and predicted matrices

The complete 5×5 measured ΔCE and predicted $-K$ matrices for all four settings are released in the results files (**results/interference*/{measured,predicted_kernel}.json**). The heterogeneous setting is visualized in Figure 2; the remaining settings follow the same block structure (large negative diagonal, retrieval capabilities weakly coupled). Accuracy deltas (Δacc) are recorded alongside ΔCE but are near zero for the already-saturated retrieval families and noisy for the rest at $n_{\text{eval}} = 30$, which is why ΔCE is the primary outcome.